

Benjamin Abensur

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EDUCATION

EPFL — Swiss Federal Institute of Technology in Lausanne

Expected Graduation: 2028

Bachelor of Science in Communication Systems

- Relevant Coursework: Introduction to Machine Learning (5.75/6), Algorithms I (5.5/6), Linear Algebra (5.5/6), Probability & Statistics, Numerical Methods for Visual Computing & ML, Computer Architecture, Computer Systems, Responsible Software

WORK EXPERIENCE

Biosynex, AI Engineer (part-time); Remote (Euronext-listed rapid diagnostics)

Oct 2026 – Present

- Building a RAG copilot over regulatory and quality documents and AI agents for the executive team.
- Training the executive team on applied AI, agentic workflows and automation pipelines.

EPFL, Teaching Assistant, Mechanics (PHYS-101) and Linear Algebra (MATH-111)

Sep 2026 – Present

- Selected on academic merit for a Mechanics course of **1,000+** students (CS, Chemistry, EE); lead a weekly 2-hour exercise session and answer questions on the course's Ed forum.

BABOO, Founder & Event Producer; Lausanne, Switzerland

2025

- Produced 3 club nights end to end that brought in **CHF 10,000**, one at Noche Club with an internationally touring DJ.

NanoSynex (Technion medtech spin-off), AI & Automation (project-based), reporting to the CEO

Remote · Jun – Sep 2024

- Benchmarked the rapid antibiotic test against a veterinary reference laboratory for a validation study, merging both result sets by sample ID in pandas and quantifying agreement rates.
- Simulated Poisson-loaded bacterial growth curves and implemented a sequential probability ratio test to call susceptible/resistant earlier, cutting time-to-result by **20%** at a **<3%** major-error rate versus the published fixed-threshold rule (Python, NumPy).
- Deployed two offline LLMs behind a model router with shared memory, enabling analysis of confidential R&D and investor documents without any data leaving the machine.
- Automated inbox triage in Python with an LLM classifier routing email into custom priority categories (investors, grants, R&D).

Sneaker Resale Venture, Partner (team of 4)

2019 – 2021

- Secured a licence to Flare AIO, a sell-out EU sneaker bot, by automating its restock purchase; ran it on Foot Locker and Snipes drops.
- Resold the licence for about twice its cost; generated roughly **€10k** over the period.

PROJECTS

Gaming Addiction Prediction — ML from scratch, CS-233 (team of 3)

github.com/Benabens/addiction-classifier-numpy

Python, NumPy, no ML libraries

- Predicted gaming-addiction level (3 classes) and score (0–10) from gaming and mental-health data with KNN, logistic/linear regression, K-Means and an MLP with hand-written backpropagation; two written reports.
- Inverse-frequency weighted cross-entropy lifted rarest-class recall from **0 to 0.69** and macro-F1 from **0.55 to 0.76** (5-fold CV).

ReCHor — public-transport journey planner, CS-108

github.com/Benabens/ReCHor

Java, JavaFX

- Finds every Pareto-optimal journey between two Swiss stops (Connection Scan Algorithm) over real CFF timetables (~258 MB).
- Memory-mapped, bit-packed timetable storage; JavaFX UI with autocomplete and iCalendar export; 44 classes, ~3,600 lines of tests.

ICoop — two-player cooperative 2D game, CS-107

github.com/Benabens/ICoop

Java

- Fire-and-water co-op game with a boss fight; interactions via double dispatch (visitor pattern). ~3,700 lines.

Kairo — internship search and application platform

kairocareer.app

Next.js, TypeScript, Python, PostgreSQL (Supabase, row-level security), GitHub Actions, Vercel

- Searches internships for the user: a daily pipeline crawls career boards, programme and free-credit feeds, deduplicates every opening and scores it against the user's profile, so none worth applying to is missed.
- Tracks every deadline, tailors the CV to each offer, flags what to build next and pre-fills application forms; 160+ tests, CI.
- Contact CRM turning freeform notes into structured contacts, then linking each opening to contacts connected to that employer (even a friend's cousin) and prompting who to follow up with.

Cortex — exam-preparation platform

cortexexam.com · GitHub

Next.js, TypeScript, PostgreSQL, Drizzle, LaTeX, Railway

- Learns each course's exam format from past papers, weights every topic by how heavily it was examined, schedules spaced revision on weak points, and generates faithful practice exams via a multi-pass LLM pipeline; used it to prepare my own exams in 3 EPFL courses.
- Answers accepted only if verified by sandboxed code execution or symbolic checks; per-user Postgres schemas, cost caps; 330+ tests.

Other Projects: Verilog digital-logic library (42/42 self-checking testbenches); networking and UNIX-filesystem project in C (CS-202, in a pair); *VeriReason*, a PyTorch Transformer from first principles (in progress); Reed-Solomon QR-code generator (Java, CS-107).

LEADERSHIP & COMMUNITY INVOLVEMENT

Scout Summer Camp, Co-Camp Director; Mévouillon, France

Jul 2025

- Co-directed a one-month camp for **80** children from two local groups (Nice, Strasbourg); led a team of **15** counsellors, promoted team leads and wrote the camp's full educational project.
- Co-managed the **€80k** camp budget, running the €18k food line myself (local farmers, Intermarché); built the site at pre-camp: tents, dry toilets, wooden tables; organised outings and explorations.

EPFL Linear Algebra Bootcamp (MATH-111), Co-instructor; Lausanne, Switzerland

Nov 2025

- Gave the lectures and wrote the exercise sets of a paid one-week bootcamp for **30** first-year students (mid-semester break).

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, Verilog, x86/RISC-V assembly

ML & AI: PyTorch, NumPy, pandas, matplotlib, Ollama

Web & Systems: Next.js, React, Node.js, REST APIs, PostgreSQL, Supabase, web scraping, GitHub Actions, Linux, POSIX sockets, Git

Spoken: French (native), English (fluent), Hebrew (basic), Spanish (basic)